🗺️ The Winds — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Chapter 41_The_Winds

Topic Index

Coriolis Force — Properties

- Coriolis force **wind velocity ke saath badhta hai** (Statement 1 TRUE); **poles pe maximum, equator pe absent/zero** (Statement 2 TRUE) — N.Hemisphere mein right deflect, S.Hemisphere mein left. Equator pe zero Coriolis force ki wajah se tropical cyclones equator ke paas NAHI bante. *[Q1]*
- **Dakshin golardh mein left deflection Coriolis force** ki wajah se hota hai (Ferrel's law: N.Hemisphere mein right, S.Hemisphere mein left turn) — Earth ki axis rotation se generate hota hai. *[Q5]*

Trade Winds aur Westerlies — Ocean Temperature Effect

- Tropical zone mein **trade winds** (E→W) ki wajah se ocean ka **western section warmer** hota hai (Statement 1 TRUE); temperate zone mein **westerlies** (W→E) ki wajah se ocean ka **eastern section warmer** hota hai (Statement 2 TRUE) — dono statements sahi hain. *[Q2]*

Wind Patterns — Clockwise/Anticlockwise Misconception

- 'Wind patterns N.Hemisphere mein clockwise, S.Hemisphere mein anticlockwise' — yeh **as general statement FALSE** hai (A galat) — **Coriolis effect dono hemispheres mein wind direction control karta hai** (R TRUE) — Low-pressure ke around N.Hemisphere mein anticlockwise, S.Hemisphere mein clockwise circulation hoti hai (High-pressure ke liye reverse). *[Q3]*
- Reverse version bhi (S.Hemisphere clockwise, N.Hemisphere anticlockwise) — yeh statement bhi **as general assertion FALSE** hai, lekin Coriolis effect ka reason (R) **TRUE** hai. *[Q4]*

> ⚠️ **EXAM TRAP**
>
> • 'Wind patterns hamesha N.Hemisphere clockwise/S.Hemisphere anticlockwise' — yeh oversimplified/galat generalization hai. Actual: low-pressure/high-pressure systems ke around directions different hote hain. *[Q3,Q4]*

Roaring Forties — Southern Hemisphere Westerlies

- S.Hemisphere ki westerlies N.Hemisphere se zyada strong+persistent hain kyunki **S.Hemisphere mein land kam hai** (Statement 1 TRUE) — **Coriolis force S.Hemisphere mein zyada hone wala Statement 2 GALAT** hai. Intensity ki wajah se **'Roaring Forties' (~40°), 'Furious Fifties' (~50°), 'Shrieking Sixties' (~60°)** naam mile — yeh **Southern Hemisphere ki westerlies** hain. *[Q6,Q8]*

Geostrophic Winds

- Coriolis effect aur pressure-gradient force ek dusre ko counter-balance karein to **Geostrophic Winds** bante hain — isobars ke parallel flow karti hain, ground se upar (kam friction) milti hain, density+pressure distribution se determine hoti hain. *[Q7]*

Roaring Forties/Furious Fifties/Shrieking Sixties

- Ye winds **Southern Hemisphere** mein bahti hain (westerlies ki intensity ke basis pe named). *[Q9]*
- Roaring Forties ke baare mein: **great strength+constancy se bahti hain, direction generally NW-to-East (S.Hemisphere), overcast skies/rain/cold weather associate hoti hai** (Statements 2,3,4 TRUE) — **dono hemispheres mein uninterrupted bahne wala Statement 1 GALAT** hai (yeh sirf Southern Hemisphere ki hain). *[Q10]*
- Roaring Forties **Indian Ocean** region mein bhi bahti hain. *[Q11]*
- 'Roaring Forties' = **40°-60° South ke beech strong ocean winds** — Equator se South Pole ki taraf air displacement + Earth rotation ka combined effect. *[Q12]*

Southern Hemisphere Deflection — Earth's Rotation

- Southern Hemisphere mein wind left deflect hoti hai **Earth ki rotation** (Coriolis force) ki wajah se — N.Hemisphere mein right, S.Hemisphere mein left. *[Q13]*

Mediterranean Sea — High-Pressure se Winds

- Conventional pressure-belt circulation ke according, high-pressure area se Mediterranean Sea ki taraf bahne wali winds **Trade Winds** hain (exam-accepted answer). *[Q14]*

Seasonal Wind Reversal — Monsoon Climate

- Seasonal wind direction reversal **Monsoon climate** ki typical characteristic hai — summer mein ~6 months sea-to-land, winter mein land-to-sea. Yeh reversal hi monsoonal climate ka keynote hai. *[Q15]*

Chinook Winds — USA Central Plains

- Chinook (Snow-Eater) winds USA ke central plains mein **winter temperature badhati hain** — winter/early spring mein common, Rocky Mountains eastern slopes se Colorado(USA) se British Columbia(Canada) tak descend karti hain. Hot wind hai, snow melt karti hai, pasture land available karti hai. *[Q16]*